International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Examination, Spring 2022

Course Code: CSE 3631 Course Title: Operating Systems

*Total marks: **50*** *Time: **2** hours **30** minutes*

The figures in the right hand margin indicate full marks.
Course Outcomes and Bloom's Taxonomy Levels are mentioned in additional columns

Group A

			CO	DL
1.				
a)	What are the necessary conditions for a deadlock to occur? Name these conditions and provide an elaboration.	3	CO2	U
b)	**Consider the following snapshot of a system:**	7	CO2	Ap

	Allocation				*Request*				*Available*			
	R_1	R_2	R_3	R_4	R_1	R_2	R_3	R_4	R_1	R_2	R_3	R_4
A	1	0	1	1	0	1	0	0	0	0	0	0
B	1	1	0	0	0	1	1	0				
C	1	0	1	1	0	0	0	0				
D	0	0	0	0	1	0	1	0				

Is the system in a deadlock state? If the system is in a deadlock state, list all processes that involve in a deadlock. Show your computation step-by-step.

2.				
a)	Compare the memory organization schemes of contiguous memory allocation and paging with respect to the following issues: 1. External fragmentation 2. Internal fragmentation	2	CO3	U
b)	Given six memory partitions of 100 MB, 170 MB, 40 MB, 205 MB, 300 MB, and 185 MB (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of size 200 MB, 15 MB, 185 MB, 75 MB, 175 MB, and 80 MB (in order)? Indicate which—if any—requests cannot be satisfied. Comment on how efficiently each of the algorithms manages memory.	5	CO3	An
c)	Consider a logical address space of 2,048 pages with a 4-KB page size, mapped onto a physical memory of 512 frames. 1. How many bits are required in the logical address? 2. How many bits are required in the physical address?	3	CO1	U

Or,

2.				
a)	Define Compaction and explain why it is used?	3	CO1	U
b)	With the help of an example explain the paging concept.	3	CO3	An
c)	Given memory holes (i.e., unused memory blocks) of 100K, 500K, 200K, 300K and 600K (in address order), how would each of the first-fit, best-fit and worst-fit algorithms allocate memory requests for 290K, 420K, 110K and 350K (in this order)?	4	CO3	Ap

Group B

3.				
a)	Under what circumstances do page faults occur? Describe the actions taken by the operating system when a page fault occurs.	3	CO1	U

b) Consider the following page reference string: 7 CO3 Ap

7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 7, 1, 0, 5, 4, 6, 2, 3, 0, 1.

Assuming demand paging with three frames, how many page faults would occur for the following replacement algorithms? Compare this two algorithms with the benchmark optimal page replace algorithm.

- LRU replacement
- FIFO replacement

4.

a) Illustrate how File-Allocation Table works. 2 CO1 U

b) Consider a file currently consisting of 100 blocks. Assume that the file-control block (and the index block, in the case of indexed allocation) is already in memory. Calculate how many disk I/O operations are required for contiguous, linked, and indexed (single-level) allocation strategies, if, for one block, the following conditions hold. In the contiguous-allocation case, assume that there is no room to grow at the beginning but there is room to grow at the end. Also assume that the block information to be added is stored in memory. 5 CO3 An

1. The block is added at the beginning.
2. The block is added in the middle.
3. The block is added at the end.

c) Explain the paging hardware with TLB. 3 CO1 U

5.

a) Explain the terms '**WORMS**' and '**VIRUSES**' with reference to system threats. 3 CO5 U

OR

Analyze the principle of least privilege regarding system protection.

b) How do computer worms work? 3 CO5 An

OR

Explain the security violation categories with example.

c) Consider the following segment table: 4 CO3 Ap

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

What are the physical addresses for the following logical addresses?

i. 0,430
ii. 1,10
iii. 2,500
iv. 3,400

OR

Consider a paging hardware with a **TLB**. Assume that the entire page table and all the pages are in the physical memory. It takes 10 milliseconds to search the **TLB** and 80 milliseconds to access the physical memory. If the **TLB** hit ratio is 0.6, then calculate the effective memory access time (in milliseconds).